Light

Embedded Systems

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AQA GCSE Computer Science — **Computer Systems: Embedded Systems**. This topic covers the definition of embedded systems, how they differ from general-purpose computer systems, and real-world examples of both types. This content is assessed in **Paper 2 (Component 2): Computing concepts**.

Exam questions typically ask you to **define** what an embedded system is, **explain** the differences between embedded and non-embedded (general-purpose) systems, and **give examples** of each type. Expect questions worth 2—4 marks requiring clear definitions and appropriate examples.

Why embedded systems matter

Embedded systems are everywhere in modern life. They control the devices we use every day, from washing machines and microwaves to cars and medical equipment. Understanding embedded systems helps you:

- **Recognise computer systems in everyday objects:** Most electronic devices contain embedded computers, even if they don't look like traditional computers
- **Understand design trade-offs:** Embedded systems are designed differently from general-purpose computers because they have different requirements

- **Appreciate the scale of computing:** There are far more embedded systems in the world than traditional computers — billions of them in cars, appliances, and industrial equipment
- **Connect hardware to software:** Embedded systems show how software controls physical devices in the real world

The AQA specification requires you to understand what embedded systems are, how they differ from general-purpose computers, and be able to give examples of both types.

What you must know

This checklist covers the key concepts for embedded systems. You should be able to explain each point clearly.

- **Understand the term** embedded system.
- **Explain how** an embedded system differs from a non-embedded (general-purpose) system.
- **Give examples** of embedded systems.
- **Give examples** of non-embedded (general-purpose) systems.
- **Describe** the key characteristics of embedded systems.
- **Explain why** embedded systems are used for certain applications.

> **Specification note.** AQA requires you to understand the term "embedded system" and explain how it differs from a "non-embedded system." The specification uses both terms: "embedded" and "non-embedded." In practice, non-embedded systems are often called "general-purpose" computers. Both terms are acceptable in exams.

Toolkit

Key definitions

Term	Definition
Embedded system	A computer system built into a larger device to perform a dedicated function or set of functions
Non-embedded system	A general-purpose computer system that can run many different programs and perform a wide variety of tasks
Dedicated function	A specific task or set of tasks that a system is designed to perform
General-purpose computer	A computer designed to run many different applications chosen by the user

Embedded vs non-embedded comparison

Feature	Embedded system	Non-embedded (general-purpose) system
Purpose	Dedicated to one specific function	Can perform many different tasks
Software	Pre-programmed, usually cannot be changed by user	User can install and run different programs
User interface	Limited or none (buttons, simple display)	Full interface (keyboard, mouse, screen)
Size	Usually small, built into another device	Standalone device, visible to user
Cost	Typically low cost per unit	Higher cost
Power consumption	Usually low power	Higher power consumption
Processing power	Only what is needed for the task	More powerful to handle varied tasks

Examples at a glance

Embedded systems

- Washing machine controller
- Microwave oven
- Digital watch
- Traffic light controller
- Car engine management system
- Central heating thermostat
- TV remote control
- Fitness tracker
- Burglar alarm
- Smart doorbell

Non-embedded (general-purpose) systems

- Desktop PC
- Laptop computer
- Tablet
- Smartphone
- Games console
- Server

Note: Smartphones and tablets are borderline cases — they are general-purpose but contain many embedded subsystems.

> **Exam tip:** When asked for examples, choose clear-cut cases. A washing machine is an obvious embedded system. A laptop is an obvious general-purpose computer. Avoid borderline examples like smartphones unless the question specifically asks about them.

1. What are embedded systems?

An **embedded system** is a computer system that is built into a larger device to perform a dedicated function or specific set of functions. Unlike a general-purpose computer, an embedded system is designed to do one job and do it well.

1.1 Definition and characteristics

The key characteristics of an embedded system are:

- **Built into another device:** The computer is part of a larger product, not a standalone computer
- **Dedicated function:** Designed to perform one specific task or a limited set of related tasks
- **Pre-programmed:** The software is usually fixed and cannot be changed by the end user
- **Often invisible to the user:** Users may not realise they are using a computer system
- **Runs continuously:** Often operates 24/7 without user intervention
- **Real-time operation:** Must respond quickly to inputs from sensors or user actions

When you use a washing machine, you interact with the embedded system through buttons and a display, but you cannot install new programs or use it for other tasks like browsing the internet. The embedded system controls the water valves, motor, heating element, and timing — all dedicated to the single purpose of washing clothes.

1.2 Components of an embedded system

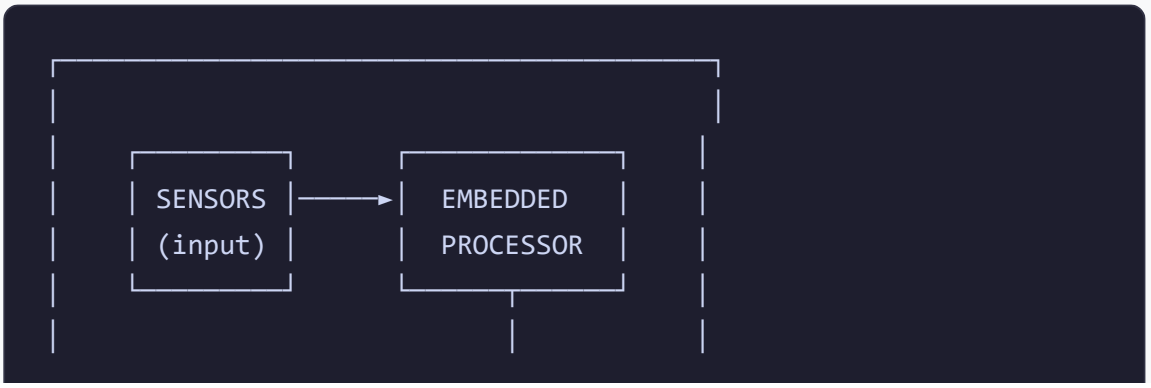
An embedded system typically contains:

- **Processor/microcontroller:** A small CPU designed for the specific task
- **Memory:** ROM/flash for the program, small amount of RAM for data
- **Input devices:** Sensors, buttons, or switches
- **Output devices:** Motors, LEDs, displays, speakers, or actuators
- **Software:** Firmware that controls the system's operation

These components are often integrated onto a single chip or small circuit board, making the system compact and cost-effective.

1.3 How embedded systems work

An embedded system follows a continuous cycle of monitoring inputs and controlling outputs:



```
│                          ▼            │
│                      ┌──────────────┐  │
│                      │  ACTUATORS   │  │
│                      │  (output)    │  │
│                      └──────────────┘  │
│                                        │
│  Example: Washing machine              │
│  • Sensor: water level detector        │
│  • Processor: decides next action      │
│  • Actuator: opens water valve         │
│                                        │
└────────────────────────────────────────┘
```

The embedded processor continuously:

1. Reads data from sensors (temperature, pressure, buttons pressed)
2. Processes the data according to its program
3. Controls outputs (motors, heaters, displays)
4. Repeats the cycle

1.4 Why use embedded systems?

Embedded systems are used instead of general-purpose computers because they offer several important advantages for dedicated applications:

- **Lower cost:** Only the necessary components are included, reducing manufacturing costs significantly. A washing machine does not need a powerful graphics card or large hard drive.
- **Smaller size:** Can fit inside compact devices where a full computer would not fit. This is essential for products like digital watches, hearing aids, and car key fobs.
- **Lower power consumption:** Important for battery-powered devices and for reducing energy costs in appliances that run for long periods.
- **Higher reliability:** Simpler systems have fewer things that can go wrong. The software is thoroughly tested for the specific application and users cannot install programs that might cause problems.
- **Faster response:** Optimised for the specific task, allowing real-time responses to sensor inputs. This is critical for safety systems like airbags and anti-lock brakes.
- **Security:** Fixed software means fewer vulnerabilities compared to systems where users can install untrusted software.
- **Longer lifespan:** Embedded systems are designed to operate for many years without updates or maintenance.

These advantages make embedded systems ideal for consumer appliances, industrial control systems, medical devices, and automotive applications where reliability, cost, and size are critical factors.

1.5 Limitations of embedded systems

While embedded systems have many advantages, they also have limitations:

- **Limited flexibility:** Cannot be reprogrammed to do different tasks
- **Difficult to update:** Fixing bugs or adding features often requires hardware replacement

- **Limited processing power:** Cannot handle complex or varied tasks
- **No user customisation:** Users cannot modify how the system works

These limitations are acceptable because embedded systems are designed for specific, well-defined tasks where flexibility is not required.

Worked Example 1 — Defining an embedded system

Question: Define the term "embedded system." [2 marks]

Model answer:
An embedded system is a computer system that is built into a larger device **(1)** to perform a dedicated/specific function **(1)**.

Exam tip: Both parts are essential for full marks: (1) built into another device, and (2) dedicated to a specific function. Missing either part loses a mark.

Worked Example 2 — Characteristics of embedded systems

Question: State two characteristics of an embedded system. [2 marks]

Model answer:
1. An embedded system performs a dedicated/specific function **(1)**.
2. The software/program is pre-programmed and cannot be changed by the user **(1)**.

Alternative answers: limited user interface, built into another device, low power consumption, runs continuously.

Exam tip: Give two distinct characteristics. Don't give two versions of the same point (e.g., "does one task" and "dedicated function" would only get one mark).

Worked Example 3 — Identifying embedded systems

Question: A modern car contains many computer systems. Give two examples of embedded systems that might be found in a car. [2 marks]

Model answer:
1. Engine management system / fuel injection controller **(1)**
2. Anti-lock braking system (ABS) controller **(1)**

Alternative answers: airbag controller, climate control system, parking sensor system, cruise control, tyre pressure monitoring system.

Exam tip: Be specific. "Computer in the car" is too vague. Name the actual system and its function.

Worked Example 4 — Why embedded systems are used

Question: Explain why an embedded system is used in a washing machine rather than a general-purpose computer. [3 marks]

Model answer:
An embedded system is cheaper than a general-purpose computer because it only contains the components needed for the washing task **(1)**. It uses less power, which reduces running costs and is better for a device that may run for long periods **(1)**. It is more reliable because the software is fixed and tested for this specific purpose, with no risk of users installing software that could cause problems **(1)**.

Exam tip: Link each advantage to the context. Don't just list "cheaper, smaller, more reliable" — explain why these matter for the specific device.

2. Embedded vs general-purpose systems

The AQA specification requires you to explain how embedded systems differ from non-embedded (general-purpose) systems. Understanding these differences is essential for exam success.

2.1 General-purpose computer systems

A **general-purpose computer** (non-embedded system) is designed to perform many different tasks. Users can install different software to use the computer for various purposes.

Key characteristics:

- **Versatile:** Can run many different programs
- **User-programmable:** Users can install and remove software
- **Full user interface:** Keyboard, mouse, large display
- **Standalone device:** A visible, separate piece of hardware
- **More powerful hardware:** Needs to handle varied, demanding tasks
- **Operating system:** Runs a full OS like Windows, macOS, or Linux

Examples of general-purpose computers:

- Desktop PCs
- Laptop computers
- Tablets
- Smartphones
- Servers

A laptop can be used for word processing, gaming, browsing the internet, editing videos, programming, and countless other tasks — all on the same device by running different

software.

2.2 Key differences

The table below summarises the main differences between embedded and general-purpose systems:

Aspect	Embedded	General-purpose
Function	Single dedicated task	Multiple varied tasks
Software	Fixed, pre-programmed	User can install programs
Visibility	Hidden inside device	Standalone, visible
User control	Limited buttons/options	Full control via OS
Hardware	Minimal, specific	Powerful, flexible
Cost	Low	Higher

2.3 Borderline cases

Some devices blur the line between embedded and general-purpose:

- **Smartphones:** General-purpose (run many apps) but contain embedded subsystems (camera controller, battery management)
- **Smart TVs:** Can run apps but primarily designed for one purpose (displaying video)
- **Games consoles:** Primarily for gaming but can run other applications

In exams, focus on clear-cut examples unless specifically asked about these borderline cases.

Visual comparison

```
EMBEDDED SYSTEM                    GENERAL-PURPOSE COMPUTER
===============                    ========================

┌──────────────────┐               ┌──────────────────────────
│   WASHING MACHINE │               │    LAPTOP COMPUTER
│                  │               │
│  ┌──────────────┐ │               │  ┌──────────────────┐
│  │   Embedded   │ │               │  │ Operating System │
│  │   Computer   │ │               │  │  (Windows/macOS) │
│  │              │ │               │  └──────────────────┘
│  │ • Fixed code │ │               │          │
│  │ • One task   │ │               │          ▼
│  │ • No OS      │ │               │  ┌────┬────┬────┐   │
│  └──────────────┘ │               │  │Word │Games│Web  │   │
│                  │               │  │Proc │     │Brwsr│   │
```

```
│  User sees: buttons │
│  and display only   │
└────────────────────┘

Cannot install new
software or change
what it does
```

```
│  └────┴────┴────┘   │
│
│  User installs any
│  software they choose
└──────────────────────

Can run thousands of
different applications
```

Worked Example 5 — Comparing embedded and general-purpose systems

Question: Explain two differences between an embedded system and a general-purpose computer system. [4 marks]

Model answer:
Difference 1: An embedded system is designed to perform a single dedicated function **(1)**, whereas a general-purpose computer can perform many different tasks by running different software **(1)**.
Difference 2: The software on an embedded system is pre-programmed and cannot be changed by the user **(1)**, whereas users can install, update, and remove software on a general-purpose computer **(1)**.

Exam tip: For comparison questions, always state both sides clearly. Use words like "whereas" or "but" to show contrast. Each difference should have a point about embedded AND a point about general-purpose.

Worked Example 6 — Identifying system types

Question: For each of the following, state whether it is an embedded system or a general-purpose computer system. [4 marks]
(a) A laptop computer
(b) A traffic light controller
(c) A desktop PC
(d) A digital camera

Model answer:
(a) General-purpose computer system **(1)**
(b) Embedded system **(1)**
(c) General-purpose computer system **(1)**

(d) Embedded system **(1)**

Exam tip: Think about whether the device can run user-installed software for different tasks. If yes, it's general-purpose. If it only does one specific job, it's embedded.

Worked Example 7 — User interface differences

Question: Describe how the user interface of an embedded system typically differs from that of a general-purpose computer. [2 marks]

Model answer:
An embedded system typically has a limited user interface with only a few buttons and possibly a small display **(1)**. A general-purpose computer has a full user interface including keyboard, mouse, and large display/monitor **(1)**.

Exam tip: "Limited interface" alone is worth one mark. Give specific examples (buttons, small display vs keyboard, mouse, monitor) to secure both marks.

Worked Example 8 — Advantages of embedded systems

Question: A company is designing a new smart thermostat for home heating. Give two reasons why they would use an embedded system rather than a general-purpose computer. [2 marks]

Model answer:
1. An embedded system would be smaller, allowing the thermostat to be compact and fit on a wall **(1)**.
2. An embedded system would be cheaper to produce, reducing the cost of each thermostat **(1)**.

Alternative answers: lower power consumption (important if battery-powered), more reliable (simpler system, less to go wrong), faster response time for temperature control.

Exam tip: Always relate your answer to the context given. The question mentions a thermostat, so explain why small size and low cost matter for that specific product.

3. Examples and applications

The AQA specification requires you to give examples of embedded and non-embedded systems. This section provides detailed examples across different categories to help you understand where embedded systems are used.

3.1 Embedded systems in the home

Kitchen appliances

- **Washing machine:** Controls water temperature, spin speed, cycle timing
- **Microwave oven:** Controls power level, cooking time, turntable motor
- **Dishwasher:** Controls water intake, detergent release, drying cycle
- **Refrigerator:** Maintains temperature, controls defrost cycle
- **Coffee machine:** Controls water heating, brewing time, milk frothing

Home electronics

- **TV remote control:** Sends infrared signals when buttons pressed
- **Digital watch:** Keeps time, controls display, alarm functions
- **Central heating thermostat:** Monitors temperature, controls boiler
- **Burglar alarm:** Monitors sensors, triggers alarm when activated
- **Smart doorbell:** Captures video, detects motion, sends alerts

Case study: Washing machine embedded system

A washing machine is an excellent example of an embedded system. Here is how it works:

Inputs (sensors and user controls):

- User buttons/dial to select wash program
- Door lock sensor (is the door closed?)
- Water level sensor (is there enough water?)
- Temperature sensor (is the water hot enough?)
- Balance sensor (is the load evenly distributed?)

Outputs (actuators):

- Water inlet valve (to fill with water)
- Drain pump (to empty water)
- Heating element (to heat water)
- Motor (to rotate the drum)
- Door lock mechanism
- Display panel (to show status)

The embedded processor runs through the wash program, reading sensor data and controlling actuators at each stage. For example, during the wash cycle it fills with water until the level sensor indicates "full," heats the water until the temperature sensor shows the correct temperature, then rotates the drum for the programmed time. The user cannot install different software or use the washing machine for anything other than washing clothes.

3.2 Embedded systems in vehicles

Modern cars contain dozens of embedded systems working together. A typical modern vehicle may have 50 to 100 embedded processors controlling everything from the engine to the entertainment system. These systems communicate with each other through internal networks within the car.

System	Function	Sensors/Inputs	Outputs/Actions
Engine management	Controls fuel injection and ignition timing for optimal performance and fuel efficiency	Temperature, airflow, throttle position, oxygen level	Fuel injectors, spark timing
Anti-lock brakes (ABS)	Prevents wheels locking during hard braking, maintaining steering control	Wheel speed sensors	Brake pressure valves
Airbag controller	Deploys airbags within milliseconds of detecting a crash	Accelerometers (impact sensors)	Airbag deployment
Parking sensors	Warns of obstacles when reversing to prevent collisions	Ultrasonic distance sensors	Beeping sounds, display
Cruise control	Maintains constant speed without driver input on the accelerator	Speed sensor, user controls	Throttle control
Climate control	Maintains cabin temperature at the level set by the user	Temperature sensors, user settings	Fans, heater, air conditioning

These automotive embedded systems must be extremely reliable because failures could be dangerous. They are designed to operate in harsh conditions including extreme temperatures, vibration, and electromagnetic interference.

3.3 Embedded systems in public infrastructure

Transport

- Traffic light controllers
- Railway signalling systems
- Speed cameras
- Ticket machines
- Barrier controls

Utilities

- Smart electricity meters
- Water pump controllers
- Street lighting systems
- Lift/elevator controllers
- HVAC systems

Healthcare

- Heart rate monitors
- Insulin pumps
- MRI scanner controllers
- Blood pressure monitors
- Pacemakers

3.4 Wearable embedded systems

Many wearable devices contain embedded systems:

- **Fitness trackers:** Monitor steps, heart rate, sleep patterns
- **Smartwatches:** Display notifications, track activity (borderline — can run apps)
- **Hearing aids:** Process and amplify sound
- **Medical monitors:** Track vital signs for patients

> **Important distinction:** A basic fitness tracker that only counts steps and shows the time is clearly an embedded system. A smartwatch that can install apps, browse the internet, and make phone calls is closer to a general-purpose device. In exams, choose clear examples like basic fitness trackers.

3.5 Summary of general-purpose computers

For comparison, here are the main types of general-purpose (non-embedded) computer systems:

Device	Why it's general-purpose
Desktop PC	Runs full operating system; user installs any software; used for many different tasks
Laptop	Portable version of desktop; same capabilities; runs varied applications
Tablet	Touchscreen device; runs apps from app store; used for browsing, games, productivity
Smartphone	Runs many different apps; used for communication, entertainment, productivity
Server	Runs different services; can be reconfigured for various purposes

Worked Example 9 — Giving examples

Question: Give two examples of embedded systems and two examples of general-purpose computer systems. [4 marks]

Model answer:
Embedded systems:
1. Washing machine controller **(1)**
2. Traffic light controller **(1)**

General-purpose computer systems:
1. Desktop PC **(1)**
2. Laptop computer **(1)**

Exam tip: Keep examples simple and clear-cut. Avoid vague answers like "computer" or "device." Be specific about what the embedded system is embedded in.

Worked Example 10 — Application context

Question: A hospital uses an embedded system to monitor a patient's heart rate. Explain why an embedded system is suitable for this purpose. [3 marks]

Model answer:
An embedded system is reliable because it performs only one dedicated task (monitoring heart rate) without the complexity of a general-purpose computer that could crash or malfunction **(1)**. It can run continuously 24/7 without user intervention, which is essential for patient monitoring **(1)**. It responds quickly in real-time to changes in heart rate, which is critical for patient safety **(1)**.

Exam tip: In healthcare contexts, emphasise reliability, continuous operation, and real-time response — these are critical for patient safety.

Worked Example 11 — Extended comparison

Question: A school office uses a desktop computer. A classroom uses an interactive whiteboard with an embedded system. Compare these two systems in terms of their purpose and flexibility. [4 marks]

Model answer:
Purpose: The desktop computer can be used for many purposes including word processing, spreadsheets, email, and internet browsing **(1)**. The interactive whiteboard's embedded system is designed for one purpose — displaying and interacting with educational content **(1)**.
Flexibility: The desktop computer is flexible because the user can install different software to change what it does **(1)**. The interactive whiteboard is less flexible because its software is pre-programmed and the user cannot install new programs on it **(1)**.

Exam tip: When comparing specific devices, address each aspect mentioned in the question (here: purpose AND flexibility) separately.

Worked Example 12 — Explaining the choice of system type

Question: A company is designing a new product that will control the temperature in a greenhouse. Discuss whether they should use an embedded system or a general-purpose computer for this product. [6 marks]

Model answer:
Arguments for an embedded system:
An embedded system would be more cost-effective because it only needs the components required for temperature monitoring and control, not a full computer **(1)**. It would be more reliable because it performs a single dedicated task with fixed software that has been tested for this specific purpose **(1)**. It would use less power, which is important if the greenhouse is in a remote location or needs to run continuously **(1)**.
Arguments for a general-purpose computer:
A general-purpose computer could be updated more easily if the requirements change, such as adding new features or connecting to other systems **(1)**. It could provide more detailed data analysis and reporting capabilities **(1)**.
Conclusion:
For a dedicated greenhouse controller, an embedded system is the better choice because cost, reliability, and low power consumption are more important than flexibility for this application **(1)**.

Exam tip: For "discuss" questions worth 6 marks, present arguments for both sides and reach a justified conclusion. Use the context (greenhouse) to explain why certain factors matter more.

Worked examples summary

The following table summarises the 12 worked examples covered in this topic, organised by section and exam skill.

Example	Focus	Marks	Key skill
1	Defining an embedded system	2	Clear, two-part definition
2	Characteristics of embedded systems	2	State distinct characteristics
3	Identifying embedded systems in cars	2	Give specific, contextual examples

Example	Focus	Marks	Key skill
4	Why embedded systems are used	3	Explain advantages with context
5	Comparing embedded and general-purpose	4	Two-sided comparison
6	Identifying system types	4	Classify devices correctly
7	User interface differences	2	Describe specific interface features
8	Advantages for specific product	2	Apply knowledge to context
9	Giving examples of both types	4	Clear, appropriate examples
10	Application in healthcare	3	Explain suitability for context
11	Extended comparison of devices	4	Compare purpose and flexibility
12	Discussing choice of system type	6	Balanced discussion with conclusion

Quick reference

Key definitions (mark-scheme style)

Term	Definition
Embedded system	A computer system built into a larger device to perform a dedicated function
Non-embedded system	A general-purpose computer that can run different programs to perform varied tasks
Dedicated function	A specific task that a system is designed to perform
General-purpose computer	A computer designed to run many different applications selected by the user
Firmware	Software that is permanently stored in an embedded system

Characteristics comparison

Characteristic	Embedded	General-purpose
Function	Single/dedicated	Multiple/varied
Software	Pre-programmed/fixed	User-installable

Characteristic	Embedded	General-purpose
Interface	Limited/buttons	Full (keyboard, mouse, screen)
Visibility	Hidden in device	Standalone
Cost	Lower	Higher
Power	Low consumption	Higher consumption
Size	Small/compact	Larger

Common examples

Embedded systems

- Washing machine
- Microwave oven
- Traffic lights
- Car engine management
- Digital watch
- Central heating thermostat
- Burglar alarm
- Fitness tracker
- TV remote control
- Lift/elevator controller

General-purpose systems

- Desktop PC
- Laptop
- Tablet
- Smartphone
- Server
- Games console (borderline)

Typical pitfalls

Pitfall 1 — Incomplete definition.
Wrong: "An embedded system is a small computer."
Right: "An embedded system is a computer system built into a larger device to perform a dedicated function."

Pitfall 2 — Vague examples.
Wrong: "A computer in a device."
Right: "A washing machine controller" or "traffic light controller."

Pitfall 3 — Confusing "dedicated" with "important."
Wrong: "Embedded systems are for important tasks."

Right: "Embedded systems perform dedicated/specific tasks" — one job, not multiple jobs.

Pitfall 4 — One-sided comparisons.
Wrong: "Embedded systems are dedicated to one task."
Right: "Embedded systems are dedicated to one task, whereas general-purpose computers can perform many different tasks."

Pitfall 5 — Saying embedded systems have no software.
Wrong: "Embedded systems don't have software."
Right: "Embedded systems have pre-programmed software (firmware) that cannot be changed by the user."

Pitfall 6 — Using borderline examples.
Wrong: Using a smartphone as an example of an embedded system.
Right: Use clear examples like washing machines, traffic lights, or digital watches.

Pitfall 7 — Forgetting to explain advantages in context.
Wrong: "Embedded systems are cheaper."
Right: "Embedded systems are cheaper because they only contain the components needed for the specific task."

Pitfall 8 — Thinking embedded means "small."
Wrong: "Embedded systems are always tiny."
Right: Size is not the defining feature. An industrial machine controller can be embedded but not small. The key feature is dedicated function.

Pitfall 9 — Confusing embedded with "built-in."
Wrong: "A built-in webcam is an embedded system."
Right: A webcam is a peripheral. An embedded system is a complete computer performing a dedicated function.

Pitfall 10 — Not stating both parts of comparison questions.
Wrong: Only describing embedded systems in a comparison question.
Right: Describe both embedded AND general-purpose to show understanding of the differences.

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